Local degree and twist bounds for circle-tree volume limits

Abstract

We prove a volume limit for the original reduced invariant of a fixed ordered tree of crossing circles from a finite condition at its vertices. The condition allows a different real-decay bound on each directed edge and imposes no sum over the tree. A simple consequence is that at every vertex either the degree is at least 2^{28} or the absolute signed crossing power is at least 2^{27}. Large-degree vertices may retain zero or small powers. The proof evaluates the full finite contraction recursively, dividing each subtree polynomial by its exact value at zero. A conditional complex kernel estimate, a one-variable Fourier calculation with all endpoint currents, and a fresh outgoing bound close the induction without a depth-dependent exponential loss. The leading coefficient is nonzero through both level parities. The classical circle-tree theorem identifies the critical action with the volume of the original hyperbolic knot complement. We retain the microscopic main-meridian scalar and give its normalized leading profile.

1 The fixed diagrams and the conclusion

A crossing circle bounds a disk punctured twice by the main strands. Its signed crossing power is measured in that disk's original frame. Fix a finite ordered tree $\mathcal{T} = (V, E)$, with at least one edge, and a crossing-circle tangle P_v at every vertex. The literal rooted construction is

$$W_v = \left(\prod_{w \in I_v^-} \mathsf{T}(W_w) \right) P_v \left(\prod_{w \in I_v^+} \mathsf{T}(W_w) \right).$$

The lists have their specified orders. Each turn rotates the whole tangle and its disk frame. There are no bare product junctions. Draw the prescribed terminal-root cylindrical closure with its actual strand orientations and one reduced cut on a main strand. Require that endpoint tracing gives one main component and no component already closed inside an intermediate tangle. These are the diagram hypotheses of [4, Section 1]; the abstract tree alone does not imply them.

Write $d_v \geq 1$ for the graph degree and $n_v \in \mathbb{Z}$ for the physical signed crossing power. The diagram and every power are fixed before the integer level $N \geq 3$ tends to infinity. Put

$$q_N = e^{\pi \mathrm{i}/N}, \qquad d = N - 1, \qquad M = q_N^s, \qquad |s| \leq R < 1.$$

The level degree d is distinct from d_v. Let $F_N(M)$ be the reduced zero-framed ADO invariant in the convention of [1], with unknot value one. At $M = 1$ it is the reduced Kashaev invariant of the same knot $K_{\mathbf{n}}$. The cut, orientation and framing are retained in every finite contraction below.

Choose any vertex o as the final vertex for evaluating that contraction. Section 2 gives a finite criterion in d_v, n_v, o and numbers $0 < c_v < 2^{-16}$ attached to the nonroot vertices. The number c_v is a chosen bound for the normalized polynomial sent from v to its parent. Its definition uses no unknown invariant or critical value.

For the action, let

$$J(x)=\int_0^x \log(2\sin \pi u)\,\mathrm{d}u, \qquad g(x)=J(2x)-4J(x), \qquad G_0=g(1/4).$$

The even holomorphic edge phase near $(0,0)$ is specified by

$$\psi(0,0)=2G_0, \qquad \psi_t(t,u)=-\frac{1}{\pi}\operatorname{arsinh}\bigl(\tan(t/2)\cos(u/2)\bigr), \qquad \psi_u(t,u)=\psi_t(u,t). \tag{1.1}$$

Branches continue from zero; the mixed derivatives agree. Put

$$\Psi(\boldsymbol{\tau})=\sum_{vw\in E}\psi(\tau_v,\tau_w), \qquad \mathcal{C}=\operatorname{stat}_{\boldsymbol{\tau}}\left\{\Psi(\boldsymbol{\tau})+\frac{\mathrm{i}}{4\pi}\sum_v n_v\tau_v^2-\mathrm{i}\sum_v \tau_v\right\}.$$

The proof constructs the small critical point used in this notation.

Theorem 1.1. *Suppose the chosen root and bounds c_v satisfy the directed criterion (2.3), (2.8), (2.9). There is a constant $D_{\mathcal{T},\mathbf{n}}\neq 0$, independent of level parity, such that*

$$F_N(1)=\Omega_N N^{3/2} D_{\mathcal{T},\mathbf{n}} e^{N\mathcal{C}}(1+O(N^{-1})), \qquad \Omega_N=q_N^{d^2(2w(D)+\sum_v n_v)/4}. \tag{1.2}$$

Here $w(D)$ is the original oriented writhe and $|\Omega_N|=1$. The original knot complement is hyperbolic, and

$$\lim_{N\to\infty}\frac{2\pi}{N}\log|F_N(1)|=2\pi\Re\mathcal{C}=\operatorname{Vol}(S^3\setminus K_{\mathbf{n}}). \tag{1.3}$$

Moreover, with

$$H_N(s)=\frac{\sin \pi s}{N\sin(\pi s/N)}, \qquad A_N(s)=q_N^{(-w(D)-\sum_v n_v)s^2/2}, \tag{1.4}$$

where $H_N(0)=1$, one has uniformly for $|s|\le R<1$

$$\frac{F_N(q_N^s)}{A_N(s)H_N(s)F_N(1)}=1+O_R(N^{-1}). \tag{1.5}$$

These values are nonzero for all sufficiently large N, and the volume limit holds uniformly with $F_N(q_N^s)$ in place of $F_N(1)$.

All implied constants and starting levels may depend on the fixed tree, its powers and R. The numerical curvature bounds and the directed criterion do not acquire an extra constant from the depth of the tree.

Corollary 1.2. *The theorem holds if, independently at each vertex,*

$$\boxed{\,d_v\ge 2^{28} \quad \text{or} \quad |n_v|\ge 2^{27}.\,}$$

It also holds under the uniform local condition $\max_v d_v/(d_v^2+n_v^2)\le 2^{-120}$.

The proof of the corollary is in Section 9; it covers all degrees symbolically. The tree, its degrees and all powers remain fixed. A high-degree vertex may have $n_v=0$. Neither condition is a hypothesis that the sum of $d_v/(d_v^2+n_v^2)$ is small.

Inputs and proof boundary. The exact finite inputs are [1, Proposition 6.1, Theorem 7.1 and Proposition 8.1]: full crossing polynomials and traces, original typed tree contraction, and scalar ledger. The unconditional estimates in [2, Lemma 2.1, Sections 3–4 and Section 6.1] supply polynomial norm comparisons, the complex edge sum and the exact one-variable Fourier conversion. Their proofs use no summed criterion. Canonical factorial interpolation is [3, Lemma 2.1]. The geometric input is [4, Theorem 1.1], whose degree-and-twist threshold follows from our criterion. We prove the conditional normalization, directed bounds, complete recursive induction and noncancellation here. No all-orders expansion, longitude factor, larger strict Conway grammar or fixed M-neighborhood is asserted.

2 A finite criterion on directed edges

The root in this section is an order for evaluating the closed finite contraction. It need not be the terminal root used to draw the original knot. A nonroot vertex sends one polynomial to its parent; the last vertex sends none. The following constants and inequalities reflect that difference.

Put

$$\begin{gathered} a_0 = \frac{1}{2\pi}, \qquad k_0 = 2^{-16}, \qquad R_0 = \frac{1}{8}, \qquad b_0 = 2^{-11}, \\ r_0 = \frac{1}{16}, \qquad t_0 = 2^{-14}, \qquad \chi = 1 - \cos r_0. \end{gathered} \tag{2.1}$$

The parent-variable rectangle is

$$\mathcal{R} = \{u : |\Re u| \le R_0,\ |\Im u| \le b_0\}. \tag{2.2}$$

Choose any vertex as root and direct all edges toward it. For every nonroot vertex choose a number

$$0 < c_v < k_0, \qquad \delta_v = \min\left\{\frac{b_0}{2}, \frac{c_v\chi}{4(1+c_v)}\right\}. \tag{2.3}$$

These are parameters of the criterion. They can differ at every edge. They are not asymptotic quantities to be inferred from the invariant.

Write $q_v = 1$ at a nonroot vertex and $q_v = 0$ at the root. If $\mathrm{Ch}(v)$ is the set of children, define

$$\begin{gathered} a_v = \frac{d_v}{2\pi}, \qquad b_v = \frac{n_v}{2\pi}, \qquad D_v = a_v - ib_v, \\ A_v = \frac{2}{\pi^2}\left(q_v k_0 + \frac{1}{2}\sum_{w\in\mathrm{Ch}(v)} c_w\right), \quad B_v = 16q_v + |\mathrm{Ch}(v)|, \quad C_v = B_v + \frac{b_v^2}{4A_v}. \end{gathered} \tag{2.4}$$

There is at least one incidence at every vertex, so A_v, B_v are positive. Let

$$\delta_v^{\mathrm{in}} = \min(\{b_0\} \cup \{\delta_w : w \in \mathrm{Ch}(v)\}) \tag{2.5}$$

and set

$$y_v = \min\left\{\frac{\delta_v^{\mathrm{in}}}{2}, \frac{t_0|D_v|}{4C_v}, \frac{A_v\pi^2}{8(\pi|b_v| + t_0|D_v|)}, \sqrt{\frac{A_v\pi^2}{8B_v}}\right\}. \tag{2.6}$$

Finally define two action quantities

$$\beta_v = \frac{210}{361}\frac{a_v}{|D_v|^2}, \qquad \eta_v = \min\left\{\frac{1}{2}t_0|D_v|y_v, \frac{1}{2}A_v\pi^2, A_v r_0^2, \frac{1}{8}a_v r_0^2\right\}. \tag{2.7}$$

The criterion consists of the strict inequalities

$$\boxed{t_0|D_v| \geq 2, \qquad \beta_v < \frac{\eta_v}{2} \quad \text{at every vertex,}} \tag{2.8}$$

and, only at nonroot vertices,

$$\boxed{\beta_v < (k_0 - c_v)\chi.} \tag{2.9}$$

It is a finite elementary test in the degrees, signed powers, root and positive numbers c_v. It uses no value of an unknown quantum invariant and no assumed dominance of a global saddle. Its quantities are local: A_v and the available strip depend on the child budgets, while the exact child scalar will be divided out rather than estimated.

We say that the chosen root and numbers c_v satisfy the directed criterion when (2.3), (2.8) and (2.9) hold. No numerical estimate of an invariant enters this definition. The number β_v bounds the action lost in evaluating just the current trace after exact child constants are divided out. The four entries of η_v pay, respectively, for the infinite charge tail, the original endpoints, localization to a real interval, and its vertical connecting paths. The extra comparison (2.9) is used only to produce an outgoing real bound.

3 The full kernel with a fixed parent angle

The fixed Newton polynomials are

$$\begin{gathered} d = N-1, \qquad c_j = 2\cos((2j+1)\pi/N), \qquad S_k = \prod_{r=1}^{k} 2\sin(\pi r/N), \\ e_k(z) = S_k^{-2}\prod_{j<k}(c_j - z), \qquad 0 \leq k < N. \end{gathered} \tag{3.1}$$

Let $\mathcal{T}_N(2\cos\theta) = 2\cos(N\theta)$, and put $P_\sigma(z) = \mathcal{T}_N(z) + 2\cos(2\pi\sigma)$. The normalized polynomial trace is

$$\phi_\sigma(F) = (-1)^d N^2 [z^d] \operatorname{rem}_{P_\sigma} F, \qquad \phi_\sigma(e_i e_j) = \mathbf{1}_{i+j=d}. \tag{3.2}$$

The remainder has degree less than N. This is the full polynomial functional, including its derivative values when roots coalesce. [1, Theorem 7.1] retains the paired factors

$$W_k(s) = \prod_{r=1}^{k}\left(1 - \frac{\sin^2(\pi s/N)}{\sin^2(\pi r/N)}\right). \tag{3.3}$$

After lowering the pairing (3.2), a physical turn gives the equal-index weight $\gamma_e(s)H_N(s)/W_k(s)$. The factor γ_e is the original scalar prescribed by that edge's orientation type. It stays in the exact scalar ledger.

For the edge action write

$$\begin{gathered} J(x) = \int_0^x \log(2\sin\pi v)\,dv, \qquad g(x) = J(2x) - 4J(x), \qquad G_0 = g(1/4), \\ b(x,t) = \int_0^x \log(2\cos(2\pi v) + 2\cos t)\,dv - 2J(x), \\ \psi(t,u) = \operatorname{stat}_x[b(x,t) + b(x,u)], \qquad \psi(0,0) = 2G_0. \end{gathered} \tag{3.4}$$

The logarithms and stationary branch are continued from $x = 1/4, t = u = 0$. The canonical factorial interpolation is [3, Lemma 2.1]. Its midpoint application and the finite complex edge contour are proved in [2, Section 3]. The full polynomial and repeated-root bounds are [2, Lemma 2.1]. These statements precede the summed hypothesis in their proofs and hold independently of it. Use the exact full kernel

$$K_N(t,u;s) = \sum_{k=0}^{N-1} e_k(-2\cos t)e_k(-2\cos u)\frac{H_N(s)}{W_k(s)}. \tag{3.5}$$

The definitions of e_k, W_k and the fixed anti-diagonal trace metric are unchanged. It is a polynomial of degree at most $N-1$ in each of $-2\cos t$ and $-2\cos u$, hence has trigonometric degree at most $N-1$ in either angle. The paired factors and their inverses are bounded for fixed $|s| \le R < 1$. Every inequality below permits a polynomial factor in N.

The full-kernel real estimate is

$$|K_N(t,u;s)| \le C_R N^{A_R} e^{N\{2G_0 - k_0[(1-\cos t)+(1-\cos u)]\}}, \qquad t,u \in [-\pi,\pi]. \tag{3.6}$$

To verify this, split the sum at $I = [1/5, 3/10]$. On I, [2, equation (4.1)] gives angular loss $\sin^2(t/2)/160$ at each end. The positive zero-angle sum has rate $2G_0$. Outside I, the full polynomial bound has loss $\eta_K = \pi/200$, including high-index derivatives. Since $4k_0 < \eta_K$ and $2k_0 < 1/160$, both parts obey (3.6). The bounded paired insertion changes only its polynomial prefactor.

The conditional estimate needed here is

$$|K_N(t+iy,u;s)| \le C_R N^{A_R} e^{N\{\Re\psi(0,u) - k_0(1-\cos t)+16y^2\}}, \quad u \in \mathcal{R}, \quad t \in [-\pi,\pi], \quad |y| \le b_0. \tag{3.7}$$

Thus the parent compact is a long, thin rectangle, rather than the old disk of radius 2^{-12}. The constant term is the actual conditional phase $\psi(0,u)$, not its upper bound $2G_0 + O(|u|^2)$.

Here is a proof from the original finite kernel. Split the indices into $I = [1/5, 3/10]$ and its complement. The complementary full polynomial, by [2, Lemma 2.1 and equation (2.7)], has the single-edge loss $\eta_K = \pi/200$, including high indices and their complete derivatives. Its ellipse bound is

$$C_R N^{A_R} e^{N\{2G_0 - \eta_K + |y| + |\Im u|\}}. \tag{3.8}$$

The edge Hessian in [2, equation (4.3)] gives, on the radius $1/4$ bidisk,

$$|\psi_{tt} + a_0| \le a_0/50, \quad |\psi_{uu} + a_0| \le a_0/50, \quad |\psi_{tu}| \le a_0/50, \quad \psi_t(0,u) = \psi_u(t,0) = 0. \tag{3.9}$$

In particular $|\psi(0,u) - 2G_0| \le |u|^2/10$. Hence (3.8) is smaller than (3.7): the sum $b_0 + b_0 + (R_0^2 + b_0^2)/10 + 2k_0$ is less than $\eta_K/2$. All these constants are independent of the tree.

For the retained indices write $u = a + ib$. Their factors at a are positive. If $r_j = \cos(2\pi(j+1/2)/N)$, then $r_j > -1/3$ and $r_j + \cos a > 253/384$. Along $|a| \le 1/8$, $|b| \le 2^{-11}$, the denominator $r_j + \cos(a+ib)$ has modulus greater than $3/5$, and

$$\left|\frac{d^2}{du^2}\log(r_j + \cos u)\right| = \left|\frac{-r_j\cos u - 1}{(r_j + \cos u)^2}\right| < 9. \tag{3.10}$$

The first derivative at real a is real. Integrating vertically therefore gives the actual finite-product estimate

$$|e_k(-2\cos(a+ib))| \le e_k(-2\cos a)e^{5Nb^2}, \qquad k/N \in I. \tag{3.11}$$

At the other end retain the original product bound

$$|e_k(-2\cos(t+iy))| \le e^2 e_k(-2) e^{N[-\alpha \sin^2(t/2)+11y^2]}, \qquad \alpha = 1/160 > 1/250. \tag{3.12}$$

The positive retained sum at $(0,a)$ is bounded by a polynomial times $\exp(N\psi(0,a))$, uniformly for real $|a| \le 1/8$. This is the positive one-edge Laplace estimate in [2, Section 3]: its unique critical wire is

$$\tan^2(\pi x) = \frac{1}{1+\tan^2(a/2)}, \tag{3.13}$$

which stays in the interior of I. Its second derivative stays negative on this fixed real compact. Equations (3.11)–(3.12), and the second bound in (3.9), consequently give

$$|K_{N,I}(t+iy,u;s)| \le C_R N^{A_R} e^{N\{\Re\psi(0,u)-\alpha\sin^2(t/2)+11y^2+6b^2\}}. \tag{3.14}$$

For $|t| \ge 1/8$,

$$6b_0^2 < \frac{\alpha}{2}\sin^2(1/16), \qquad 2k_0 < \alpha/2. \tag{3.15}$$

For example $\sin(1/16) > 1/(8\pi)$, $\pi^2 < 10$ and $\alpha > 1/250$ already prove the first inequality. This proves (3.7) in this range.

For $|t| \le 1/8$, [2, equation (3.10)] gives the exact nonzero edge asymptotic on the $1/4$ bidisk. Both u and $t+iy$ lie strictly inside that disk. Integrating the first bound of (3.9) gives (3.7) there. The original lattice currents and (3.8) are exponentially smaller. In particular the full kernel, uniformly on this smaller rectangle product, retains

$$K_N(t,u;s) = N^{-3/2} a_e(t,u) e^{N\psi(t,u)}(1+O_R(N^{-1})), \tag{3.16}$$

with the original nonzero holomorphic leading amplitude and midpoint normalization. The outside physical edge gamma is not included in a_e.

4 The exact recursion may be rooted anywhere

Lower the fixed Frobenius pairing at every vertex of the original closed source. Every physical turn then identifies equal Newton indices and retains its factor $\gamma_e H_N/W_k$. The resulting finite scalar is a sum over the original edge indices of a product of the original vertex trace tensors. Choosing a root merely specifies the order of these finite contractions. It does not move a cut through a tangle, commute open operators or change a framing.

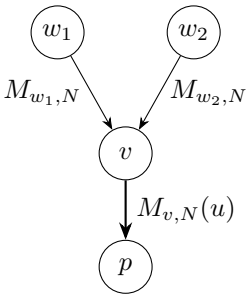
Consequently the exact recursions of [1, Theorem 7.1] hold for any chosen root:

$$M_{v,N}(u) = \gamma_{e_v}(s)\, \mathcal{D}^{(t)}_{n_v,\sigma_v,N}\left[K_N(t,u;s) \prod_{w\in \mathrm{Ch}(v)} M_{w,N}(t)\right] \tag{4.1}$$

at a nonroot vertex, where $\mathcal{D}_{n,\sigma,N}[f] = \phi_\sigma(R_\sigma^n f)$ uses the actual physical crossing polynomial and its Cartan scalar. At the root the recursion is

$$Z_N = \mathcal{D}^{(t)}_{n_o,\sigma_o,N}\left[\prod_{w\in \mathrm{Ch}(o)} M_{w,N}(t)\right] \tag{4.2}$$

at the root. The original global closing, framing and reference factors of [1, Proposition 8.1], and the original single reduced cut H_N, remain outside (4.2). Edge gammas are counted once.



$$\widehat{M}_{w,N}(t) = M_{w,N}(t)/M_{w,N}(0)$$
$$M_{v,N}(u) = \gamma_{e_v}\left(\prod_w M_{w,N}(0)\right)$$
$$\cdot\mathcal{D}_v\left[K_N(t,u)\prod_w \widehat{M}_{w,N}(t)\right]$$

At the final root: omit $K_N(t,u)$ and γ_{e_v}. There is no outgoing polynomial.

Figure 1: Edges point toward the vertex whose trace is evaluated next. The exact numbers $M_{w,N}(0)$ are factored out before applying that trace. The outgoing polynomial is then divided by its own exact value at zero. Only a nonroot vertex must supply an outgoing real-decay bound; the final root evaluates a scalar. This diagram describes contraction order, not an isotopy of the knot.

The exact normal value $\sigma_v = 0$ and the moving values $\pm s$ retain their actual orientation types.

Each message is an even trigonometric polynomial of degree at most $N-1$ in u, regardless of the subtree. There is no discarded-source error in (4.1)–(4.2).

5 The phase class and the variable envelopes

The inductive analytic class is

$$h_v(0) = h_v'(0) = 0, \qquad h_v(-u) = h_v(u), \qquad |h_v''(u) + a_0| \le a_0/20, \quad u \in \mathcal{R}. \tag{5.1}$$

The leading normalized amplitude $\alpha_v(u)/\alpha_v(0)$ is holomorphic and nonzero on a neighborhood of the required smaller compact. Exact subtree constants are retained:

$$\frac{M_{v,N}(u)}{M_{v,N}(0)} = \frac{\alpha_v(u)}{\alpha_v(0)} e^{Nh_v(u)}(1 + O(N^{-1})), \quad u \in \mathcal{R}. \tag{5.2}$$

As usual a slightly smaller rectangle can be used for fixed derivative estimates; the strict margins permit an initially slightly larger one.

The global assertions attached to c_v are

$$\left|\frac{M_{v,N}(t)}{M_{v,N}(0)}\right| \le CN^A e^{-Nc_v(1-\cos t)}, \quad t \in [-\pi, \pi], \tag{5.3}$$

$$\left|\frac{M_{v,N}(t+iy)}{M_{v,N}(0)}\right| \le CN^A e^{N[-(c_v/2)(1-\cos t)+y^2]}, \quad |y| \le \delta_v. \tag{5.4}$$

For canonical representatives of degree less than N, the real sup, Chebyshev and full value-and-first-derivative norms differ by polynomial factors, by [2, Lemma 2.1]. A product of degree a fixed multiple of N is first bounded in its real sup norm and then folded in the correct quotient; no equivalence with finite node data is asserted for an unreduced higher-degree polynomial. In particular the reduction and trace retain the repeated-root normals.

Here is the passage from the real bound to the strip. Write the normalized message as a Laurent polynomial $P(e^{it})$, of powers between $-m$ and m, where $m < N$. The subharmonic function $\log|z^m P(z)|$ on the unit disk is bounded by the harmonic function $\log(CN^A) -$

$Nc_v + Nc_v \Re z$. Substitute $z = e^{-y+it}$ and divide by $|z|^m$; for negative y use the reciprocal polynomial. Since $1 - e^{-|y|} \le |y|$, this gives

$$|M_{v,N}(t+iy)/M_{v,N}(0)| \le CN^A e^{N[-c_v(1-\cos t)+(1+c_v)|y|]}. \tag{5.5}$$

For $|t| \ge r_0$ use (2.3). For $|t| < r_0$ use the rectangle expansion and

$$\Re h_v(t+iy) \le -\frac{19a_0}{40}t^2 + \frac{21a_0}{40}y^2. \tag{5.6}$$

Both arguments give (5.4), without a conjectured quadratic estimate for an individual high Newton basis element.

Assume the children satisfy (5.1)–(5.4). At a nonroot vertex set

$$A_v^{\mathrm{loc}}(t,u) = \psi(t,u) + \sum_w h_w(t), \quad C_{v,\ell}(u) = \mathrm{stat}_t\{A_v^{\mathrm{loc}}(t,u) + ib_v t^2/2 - i\ell t\}. \tag{5.7}$$

At the root omit ψ and u. There are d_v incidences in t. Hence

$$A_v^{\mathrm{loc}}(t,u) = \psi(0,u) - a_v t^2/2 + E_v(t,u), \quad |E_{v,tt}|,\ |E_{v,t}/t| \le a_v/20. \tag{5.8}$$

The root constant in this formula is zero. For $0 \le \ell \le L_v = \lfloor t_0|D_v| \rfloor$ the equation $D_v t = -i\ell + E_{v,t}$ contracts in $|t| \le b_0$. Indeed $t_0 + b_0/20 < b_0$. The actual solutions lie in the smaller disk $|t| \le t_0/(19/20)$, leaving room for the translated contours.

Writing $E_{v,t} = tQ_v$, with $|Q_v| \le a_v/20$, gives $t_{v,\ell} = -\mathrm{i}\ell/(D_v - Q_v)$. Its imaginary part is $-\ell\Re(D_v - Q_v)/|D_v - Q_v|^2$. The numerator lies between $19a_v/20$ and $21a_v/20$, and the denominator between $(19|D_v|/20)^2$ and $(21|D_v|/20)^2$. Therefore

$$\frac{380}{441}\frac{\ell a_v}{|D_v|^2} \le -\Im t_{v,\ell} \le \frac{420}{361}\frac{\ell a_v}{|D_v|^2}, \tag{5.9}$$

also for complex parent u. No child phase is asserted to be real on the real axis. Its nonlinear derivative is t times a function bounded by $a_v/20$, so the real part of the denominator remains positive.

The normalized phase $h_v = C_{v,1}(u) - C_{v,1}(0)$ satisfies

$$h_v' = \psi_u(t_{v,1}, u), \qquad h_v'' = \psi_{uu} - \frac{\psi_{ut}^2}{A_{v,tt}^{\mathrm{loc}} + ib_v}. \tag{5.10}$$

Using (3.9) and $|A_{v,tt}^{\mathrm{loc}} + ib_v| \ge (19/20)|D_v| \ge (19/20)a_0$ proves (5.1), since $1/50 + 20/(19 \cdot 2500) < 1/20$. This is uniform on the rectangle. Charge ordering and the current-vertex drop are

$$\Re C_{v,\ell}(u) \le \Re C_{v,1}(u) - \frac{190}{441}\frac{a_v}{|D_v|^2}(\ell^2 - 1), \quad 0 \le \Re\psi(0,u) - \Re C_{v,1}(u) \le \beta_v. \tag{5.11}$$

At the root again replace $\psi(0,u)$ by zero.

6 Actual trace tails with the directed budgets

Divide every child by its exact nonzero value $M_{w,N}(0)$. The induction uses those complex constants exactly; it never estimates them by their complete-source magnitudes. Equations (3.7) and (5.4) imply

$$|\text{normalized trace input}(t+iy,u)| \le CN^{A}e^{N[\Re\psi(0,u)-A_vt^2+B_vy^2]}, \quad |y|\le \delta_v^{\text{in}}. \tag{6.1}$$

The root constant is zero. The factor $2/\pi^2$ in (2.4) is exactly the inequality $1-\cos t \ge 2t^2/\pi^2$ on the original interval.

Write $\eta=(-1)^d$. For the actual input, an even trigonometric polynomial of any fixed-multiple degree, the exact trace is

$$\mathcal{D}_{n,\sigma,N}[f]=\frac{\eta NC_{N,n}(\sigma)}{\sin(2\pi\sigma)}\sum_{j=0}^{N-1}\sin\tau_j\, f(\tau_j)e^{\mathrm{i}nN\tau_j^2/(4\pi)}, \quad \begin{cases}\tau_j=\dfrac{2\pi}{N}(d/2+\sigma-j),\\ C_{N,n}(\sigma)=q_N^{n(d^2/4-\sigma^2/2)}.\end{cases} \tag{6.2}$$

This is [1, equation (6.4)], initially away from colliding grids; its holomorphic limit includes every normal. To recall the exact Fourier conversion, use

$$\mathcal{K}_\sigma(t)=\frac{\mathrm{i}Ne^{\mathrm{i}Nt}}{e^{\mathrm{i}Nt}-\eta e^{2\pi\mathrm{i}\sigma}}$$

on the rectangle with sides $\pm\pi\pm\mathrm{i}y_v+2\pi\sigma/N$. The kernel has residue one at the N enclosed grid points and equals $\mathrm{i}Ne^{-Nz}/(1+e^{-Nz})$ on a vertical side $t=\pm\pi+\mathrm{i}z+2\pi\sigma/N$. Its upper and lower horizontal series are

$$-\mathrm{i}N\sum_{\ell\ge1}\eta^\ell e^{-2\pi\mathrm{i}\ell\sigma}e^{\mathrm{i}\ell Nt}, \qquad \mathrm{i}N\left(1+\sum_{\ell\ge1}\eta^\ell e^{2\pi\mathrm{i}\ell\sigma}e^{-\mathrm{i}\ell Nt}\right).$$

Move each entire term to a fixed horizontal segment, retaining its short endpoint connectors. Reflect the upper integral by $t\mapsto -t$, with the boundary orientation; $\sin t\, f(t)e^{\mathrm{i}nNt^2/(4\pi)}$ is odd. The paired coefficients are therefore

$$c_{\ell,N}(\sigma)=\eta^{\ell-1}U_{\ell-1}(\cos 2\pi\sigma), \qquad c_{1,N}=1, \tag{6.3}$$

with common prefactor $-N^2C_{N,n}/(\pi\mathrm{i})$. The zero-mode integral on the real interval vanishes by oddness, but its endpoint currents remain. These algebraic steps are also written in [2, Section 6.1]; they impose no condition on the twists or sum over vertices. In particular there is no later sum of unexamined equal-action signs.

For $\ell>L_v$, moving the single-grid rectangle by y_v costs, after maximizing the chirp term in the real variable, at most $C_vy_v^2$. Since $\ell>L_v$ implies $\ell>t_0|D_v|$, (2.6) gives a tail loss at least

$$t_0|D_v|y_v-C_vy_v^2\ge\frac{1}{2}t_0|D_v|y_v. \tag{6.4}$$

Successive charges have the additional factor $\exp(-Ny_v)$. The exact fixed-compact growth of (6.3) is absorbed for all sufficiently large N. This starting level may depend on the fixed vertex.

At an original endpoint $\pm\pi$, the chirp and finite-charge connectors cost at most

$$(\pi|b_v|+t_0|D_v|)y_v+B_vy_v^2\le A_v\pi^2/4. \tag{6.5}$$

Thus their loss is at least $A_v\pi^2/2$. This includes the original zero-mode current. Nothing is inferred by dropping the endpoint terms of the nonperiodic chirp.

For a retained charge, restrict its real integral to $[-r_0,r_0]$. Its complement loses $A_vr_0^2$ by (6.1). Translate the retained interval to $\Im t_{v,\ell}$. Its endpoints remain in the analytic rectangle. The original local Young inequality, now on this larger real interval, is

$$\Re\{A_v^{\mathrm{loc}}(t-iz,u)+ib_v(t-iz)^2/2-i\ell(t-iz)\}\le \Re\psi(0,u)-a_vt^2/8 + \left[\frac{21}{40}a_v+\frac{2b_v^2}{(14/5)a_v}\right]z^2-\ell z. \tag{6.6}$$

For $0\le z\le -\Im t_{v,\ell}$, the last line after its quadratic term is at most $-\ell z/8$, since $(420/361)\max(21/40,5/7)=300/361<7/8$. Every such connector therefore loses at least $a_vr_0^2/8$. The translated horizontal saddle has negative real quadratic part at most $-(19/20)a_v$. The physical sine factor is nonzero at $t_{v,\ell}$.

Equations (6.4)–(6.6) give an error of at most $CN^A\exp N[\Re\psi(0,u)-\eta_v]$. Nonminimal charges are their saddle terms plus this absolute error; no relative approximation is asserted at a much lower nonminimal action. Conditions (2.8) and (5.11) put the whole conditional error strictly below the first charge. Equation (6.3) gives its nonzero coefficient for both parities. This establishes (5.2) and nonvanishing of the exact $M_{v,N}(0)$.

For the child inputs, their exact analytic logarithms differ from the limiting phases by $O(N^{-1})$, with fixed derivatives on the required smaller compact. The finite product of those corrections is retained in the amplitude, or moves the saddle by $O(N^{-1})$. As the tree is fixed, it spends no new exponential budget. The leading amplitude is a product of nonzero edge, child and Gaussian factors. All N-dependent physical crossing and carrier phases remain in $M_{v,N}(0)$; its normalized leading amplitude does not conceal such phases.

The full physical trace and the first-charge approximation are holomorphic in the formal meridian variables. As in the original proof, first take parameter circles avoiding collisions and then apply the maximum principle to their whole difference. This covers every actual $\sigma_v=0,\pm s$, including normal limits. No isolated divided current is declared removable without its companions.

7 Regenerating the outgoing real bound

Lemma 2.1 of [2] compares the real sup norm, the Chebyshev norm, and the complete value-and-first-derivative norm by polynomial factors. It also bounds the physical fixed-power crossing and the full quotient trace. These estimates apply to polynomial products of degree a fixed multiple of N; that multiple can depend on the degree of this fixed vertex. For real u, (3.6), the polynomial real sup bounds of the normalized children, the actual fixed-power crossing polynomial, Chebyshev folding in the correct quotient, and the full top-coefficient functional give

$$|M_{v,N}(u)|\le CN^A\left|\prod_w M_{w,N}(0)\right|e^{N[2G_0-k_0(1-\cos u)]}. \tag{7.1}$$

This retains repeated-root derivatives and all fixed normals. Dividing by the nonzero first-charge value at zero costs at most β_v. For $|u|\ge r_0$, (2.9) absorbs that cost and yields (5.3). For $|u|<r_0$, (5.2) and (5.6) give the stronger estimate, since $c_v<k_0\ll 19a_0/20$. Equation (5.4) follows by Section 5. Thus the induction closes with the selected directed budgets, rather than a common decay constant.

There is no outgoing real bound at the final root. Imposing (2.9) there would be an unnecessary restriction. Its only normalization comparison is the conditional trace estimate (2.8).

8 Original action, geometry, and scope

Restore the exact subtree constants successively. Stationary elimination in (5.7) gives the original joint equations

$$\sum_{e\sim v}\psi_{e,\tau_v}+\frac{in_v\tau_v}{2\pi}=i. \tag{8.1}$$

The exact child constants telescope to the original critical value

$$\mathcal{C}(\mathbf{1})=\operatorname{stat}_{\boldsymbol{\tau}}\left[\sum_e\psi_e+\frac{i}{4\pi}\sum_v n_v\tau_v^2-i\sum_v\tau_v\right]. \tag{8.2}$$

Back-substitute from the final root: its first-charge angle belongs to the rectangle, so each child's selected parent value does also, and the same statement continues down the finite tree. Every restored angle has modulus at most $1/[(19/20)|D_v|]$. Thus the vector satisfying (8.1) lies inside the full simultaneous small-angular uniqueness polydisk, not just the conditional disks used during elimination. The existing contraction identifies it with the same branch as the marked positive charge path of [4, Theorem 1.1]. In fact $20/(19|D_v|)<10^{-2}$. Indeed (2.8) implies

$$\sqrt{d_v^2+n_v^2}=2\pi|D_v|\geq 2\pi\,2^{15}>630. \tag{8.3}$$

[4, Theorem 1.1] applies with no sum condition. The edge functions agree: their derivatives are (1.1) and their complete values satisfy $2G_0=v_{\mathrm{oct}}/\pi$, as in [2, Section 7]. Write $n_v=\epsilon_v-2p_v$, with $\epsilon_v\in\{-1,0,1\}$, in the original disk frame. The primitive periods supplied by the geometric theorem are

$$U_v=-2\pi\Psi_{\tau_v}-\mathrm{i}\epsilon_v\tau_v,\qquad V_v=2\mathrm{i}\tau_v.$$

Equation (8.1) therefore gives the full continued relation $U_v+p_vV_v=-2\pi\mathrm{i}$. The imported theorem proves its primitive winding, smooth metric completion, and complete main cusp; we do not infer these facts from an exponentiated equation. Its original ordered two-octahedron cut, odd seam and calibrated action identify $2\pi\Re\mathcal{C}(\mathbf{1})$ with the volume of the actual filled knot complement. This is a geometric input, not a deduction from the quantum saddle alone.

There are $|E|$ edge Gaussians $N^{-3/2}$ and $|V|$ trace Gaussians $N^{3/2}$; $|V|=|E|+1$ leaves the original $N^{3/2}$ power with nonzero amplitude. The exact first Fourier coefficient is one. The retained edge insertion has leading value one, and the scalar ledger remains precisely (1.4). At $s=0$, all edge gammas and paired insertions equal one. The exact outside scalar is

$$\Omega_N=\theta_d^{-w(D)}\prod_v C_{N,n_v}(0)=q_N^{d^2(2w(D)+\sum_v n_v)/4},\qquad \theta_d=q_N^{-d^2/2}.$$

The remaining leading coefficient is independent of N, nonzero, and the same through both parities. This proves the full leading asymptotic in Theorem 1.1. For bounded s, the first-charge coefficient is still one, the paired insertion has leading value one, and all normalized leading amplitudes are unchanged. Restoring the exact scalar ledger of [1, Proposition 8.1] gives $A_N(s)H_N(s)$. The finite product of relative $1+O_R(N^{-1})$ factors proves (1.5). Its nonvanishing follows uniformly on every fixed disk $R<1$, since A_N and H_N have no zeros there. Together with the geometric identification above this proves Theorem 1.1.

The criterion allows different strip and real-decay budgets and a different choice of final vertex. It is not the assertion that arbitrary capped short-twist islands can be substituted as black-box messages. Such an island requires a multi-boundary version of the conditional normalization proved here for an actual edge. The pointwise source estimates on the older tiny disk alone do not supply that extension.

9 Uniform local consequences and concrete constructions

The directed criterion has the following simple consequence:

$$|n_v| \geq 2^{27} \quad \text{at every vertex} \quad \Longrightarrow \quad \text{Theorem 1.1.} \tag{9.1}$$

This sufficient constant is independent of the number of vertices, their degrees and the choice of root. Here is a proof covering all degrees simultaneously. Choose $c_v = k_0/2$ at every nonroot vertex and write δ for their common value in (2.3). The elementary cosine estimate gives

$$\chi \geq \frac{r_0^2}{2} - \frac{r_0^4}{24} = 2^{-9}\frac{3071}{3072} > 2^{-10}, \qquad \delta > 2^{-30}. \tag{9.2}$$

At any vertex, including the root, put $\mathcal{R}_v = \sqrt{d_v^2 + n_v^2}$. Then

$$A_v \geq \frac{k_0 d_v}{2\pi^2}, \qquad B_v \leq 16 d_v, \qquad C_v \leq \frac{\mathcal{R}_v^2}{8k_0 d_v}. \tag{9.3}$$

For the last inequality use $16 \leq 1/(8k_0)$. Each of the four branches of the minimum in (2.6) consequently has the following lower bound, respectively:

$$\frac{\delta}{2}, \qquad \frac{t_0 k_0 d_v}{\pi \mathcal{R}_v}, \qquad \frac{k_0 d_v}{16\mathcal{R}_v}, \qquad 2^{-12}. \tag{9.4}$$

The second displayed quantity is smaller than the third and fourth. For the third bound, use $\pi|b_v| + t_0|D_v| \leq \mathcal{R}_v(1 + t_0/\pi)/2$. Thus every branch of (2.7), not just its tail branch, is bounded below by

$$\eta_v \geq \min\left\{\frac{t_0 \mathcal{R}_v \delta}{8\pi}, \frac{t_0^2 k_0 d_v}{4\pi^2}\right\}. \tag{9.5}$$

Indeed the second entry here is smaller than each of the other three entries in (2.7), by their explicit values and $t_0 \ll r_0$. On the other hand

$$\beta_v = \frac{420\pi}{361}\frac{d_v}{\mathcal{R}_v^2} < \frac{4d_v}{\mathcal{R}_v^2}. \tag{9.6}$$

For comparison with half the first entry of (9.5), it suffices that $\mathcal{R}_v^2 > 64\pi/(t_0\delta)$, since $d_v \leq \mathcal{R}_v$. The right side is less than 2^{52}, using $\pi < 4$ and (9.2). For comparison with half the second entry it suffices that $\mathcal{R}_v^2 > 32\pi^2/(t_0^2 k_0)$, whose right side is less than 2^{53}, since $\pi^2 < 10$. Both inequalities follow strictly from $\mathcal{R}_v^2 \geq n_v^2 \geq 2^{54}$. The charge-cut condition in (2.8) follows as well.

For the outgoing patch, the sharper elementary estimate is

$$\beta_v \leq \frac{210\pi}{361|n_v|} \leq \frac{330}{361}\,2^{-26} < \frac{3071}{3072}\,2^{-26} \leq \frac{k_0}{2}\chi. \tag{9.7}$$

Here $\pi < 22/7$, $d_v/(d_v^2 + n_v^2) \leq 1/(2|n_v|)$, and the displayed rational comparison prove (2.9) for every degree. This completes the corollary without a numerical scan or a threshold depending on the tree.

The same proof gives a mixed condition, chosen independently at each vertex:

$$\boxed{d_v \geq 2^{28} \quad \text{or} \quad |n_v| \geq 2^{27} \qquad \text{for every } v.} \tag{9.8}$$

Under the degree alternative, $\mathcal{R}_v^2 \geq d_v^2 \geq 2^{56}$ proves the same two tail comparisons, while

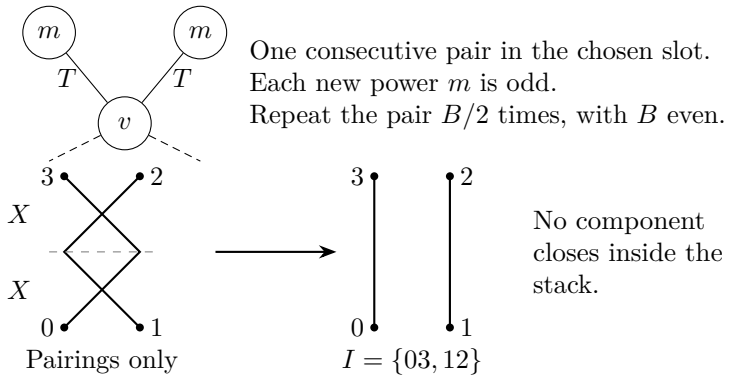


Figure 2: The upper drawing records tangle operations, not a knot projection. In the construction, each drawn pair represents $B/2$ consecutive pairs inserted in one chosen product slot. The lower pairing calculation is $X \circ X = I$, with no internal component; it preserves the original closure's component count, not its knot type.

$$\beta_v \le \frac{420\pi}{361 d_v} \le \frac{330}{361}\, 2^{-26} < \frac{k_0}{2}\chi \tag{9.9}$$

is the same outgoing patch bound. Neither alternative needs to hold uniformly over all vertices. In particular, the high-degree vertices may retain any fixed signed powers, including zero and small nonzero powers. No sum over the number of such vertices occurs.

Here is a precise extension construction illustrating that freedom. Start with any fixed ordered circle-tree core whose original endpoint pairing closes to a knot and has no internal closed component. Retain all its fixed signed powers. At every core vertex choose one of its actual product slots next to its crossing box and insert a consecutive stack of an even number $B \ge 2^{28}$ of new terminal pieces. Every new leaf has a fixed odd power satisfying

$$|n_{\mathrm{leaf}}| \ge 2^{27} + 1. \tag{9.10}$$

The physical quarter-turn of an odd two-strand braid has diagonal pairing $X = \{02, 13\}$; this pairing is invariant under the quarter-turn. An even ordered stack of these leaves has the through pairing $I = \{03, 12\}$ and creates no internal circle. Inserting it in the specified slot therefore preserves the original local endpoint pairing in that actual frame. This is a pairing calculation, not an isotopy removing the new tangles. At the original closing box the same insertion takes place inside its local ball. An added terminal leaf may then be chosen as the recursive root using the original closed-collar move with its induced frame reversal and factor order retained.

Thus the resulting diagram is again a knot, while its knot type may change. Each old vertex now satisfies the degree alternative and each new leaf satisfies the twist alternative in (9.8). The resulting actual knot complement is hyperbolic and satisfies Theorem 1.1. The number and arrangement of prescribed short-power vertices are arbitrary but fixed; the number B does not depend on their total count.

The earlier uniform local condition is also a corollary:

Corollary 9.1. *If* $\max_v d_v/(d_v^2 + n_v^2) \le 2^{-120}$*, then the conclusions of Theorem 1.1 hold.*

Proof. If both $d_v < 2^{28}$ and $|n_v| < 2^{27}$, then $d_v^2 + n_v^2 < 2^{57}$. Since $d_v \ge 1$, this would give $d_v/(d_v^2 + n_v^2) > 2^{-57} > 2^{-120}$. Thus every vertex satisfies (9.8). This proof uses the directed theorem, not a summability hypothesis over vertices. □

10 A star beyond both previous quantitative tests

Consider an ordered star rooted at its central vertex, of odd degree $d = 30{,}000{,}001$. Set its power to 2 and every terminal power to the fixed odd integer $2^{60} + 1$. Choose $c_v = k_0/2$ on every leaf.

The knot pairing can be checked in the original terminal-root syntax, even though the finite contraction is evaluated from the center. Label the boundary points lower left, lower right, upper right, upper left by $0, 1, 2, 3$. The even and odd braid pairings are respectively

$$I = \{03, 12\}, \qquad X = \{02, 13\}; \qquad T(I) = H = \{01, 23\}, \quad T(X) = X. \tag{10.1}$$

At a terminal root there are $d - 1$ other leaves. Their turned odd braids have pairing X; their ordered product has pairing I because $d - 1$ is even. The central power 2 preserves it. Its actual quarter-turn has pairing H, and stacking the root's odd braid against H still gives H, with no internal closed arc. Cylindrical closure joins 03 and 12, so it closes H into one component. All intermediate products before that last stack are two through-strand braids. There is no hidden internal component.

The criterion can be verified by rational enclosures without expanding this large tree. All leaves have the same data. The center has approximately

$$\beta_o = 1.219\,10^{-7}, \qquad \eta_o = 2.713\,10^{-7}, \qquad \frac{\beta_o}{\eta_o/2} < 0.9. \tag{10.2}$$

Every leaf has a much smaller ratio. The root does not satisfy the nonroot patch inequality for any $c > 0$, because $\beta_o > k_0\chi$. Nevertheless its exact final trace satisfies (2.8). Thus that outgoing patch must not be charged to a root that sends no message.

Moreover

$$\sum_v \frac{d_v}{d_v^2 + n_v^2} > \frac{30{,}000{,}001}{30{,}000{,}001^2 + 4} > 2^{-25}, \tag{10.3}$$

and its maximum is far above 2^{-120}. The example therefore lies beyond both published quantitative alternatives. Its size and all its powers are fixed; it is not an N-growing family. The constant sizes are coarse, and no optimality assertion is intended.

Here are rational enclosures proving the inequalities rather than inferring them from those decimal values. Put

$$\pi_- = 3, \quad \pi_+ = 22/7, \quad c = k_0/2, \quad \chi_- = r_0^2/2 - r_0^4/24, \quad \delta_- = \frac{c\chi_-}{4(1+c)}. \tag{10.4}$$

At the central vertex use the conservative height $y_* = \delta_-/2$. It is at most each branch of (2.6). For example, with

$$D_- = \frac{d}{2\pi_+}, \quad D_+ = \frac{d+1}{2\pi_-}, \quad A_- = \frac{dc}{\pi_+^2}, \quad C_+ = d + \frac{\pi_-^{-2}}{4A_-}, \tag{10.5}$$

the three non-strip comparisons are

$$y_* < \frac{t_0 D_-}{4C_+}, \qquad y_* < \frac{dc}{8(1 + t_0 D_+)}, \qquad y_*^2 < \frac{c}{8}. \tag{10.6}$$

The strip comparison follows from $\delta \geq \delta_-$. Every quantity in (10.4)–(10.6) is rational. Direct cross-multiplication also gives

$$\frac{2[420\pi_+ d/(361(d^2+4))]}{t_0 D_- y_*/2} < \frac{9}{10}, \qquad 2\frac{420\pi_+ d}{361(d^2+4)} < \min\left\{\frac{dc}{2},\ A_- r_0^2,\ \frac{D_- r_0^2}{8}\right\}. \tag{10.7}$$

These compare the upper bound for twice the first-charge drop with all four lower bounds for η_o. The charge cutoff is greater than two.

For a leaf with $n = 2^{60}+1$, take $y_* = t_0 k_0/(2n)$. The same direct rational comparisons, using $D_- = n/(2\pi_+)$, $D_+ = (n+1)/(2\pi_-)$, $A_- = 2k_0/\pi_+^2$, and $C_+ = 16 + [n/(2\pi_-)]^2/(4A_-)$, put this height below each branch of (2.6). They give a lower bound greater than 2^{-49} for every entry of (2.7), while $\beta_v < 2^{-117}$. The outgoing patch follows from $\beta_v < (k_0/2)\chi_-$. This proves the leaf inequalities.

11 Finite reproduction and limitations

From the repository root run

```
python3 papers/directed-tree-asymptotics/check.py \\
    --output /tmp/directed-tree-check.json
```

The record pins its source files. It reuses the exact rational budget and endpoint-pairing calculations, and the finite comparison of quotient messages with full root-grid and first-normal contractions. The examples in that comparison use formal independent vertex parameters; this tests the finite algebra, not the existence of additional diagrams. The checker also verifies the elementary implication from the older local condition and includes negative controls for a lost root exemption and an odd rather than even inserted stack. Finite checks neither prove the uniform analytic induction nor replace the imported geometric theorem.

References

[1] *Exact Newton calculus for tangles with a moving meridian*, companion paper, Propositions 6.1 and 8.1, Theorem 7.1. https://sashakolpakov.github.io/murakamis_dream/papers/meridian-tangles.pdf.

[2] *Degree and twist estimates for circle-tree invariants*, companion paper, Lemma 2.1 and Sections 3, 4 and 6.1. https://sashakolpakov.github.io/murakamis_dream/papers/circle-tree-asymptotics.pdf.

[3] *Holomorphic source asymptotics and volume limits for Conway closures*, companion paper, Lemma 2.1. https://sashakolpakov.github.io/murakamis_dream/papers/conway-asymptotics.pdf.

[4] *Positive octahedra and primitive fillings for trees of crossing circles*, companion paper, Theorem 1.1. https://sashakolpakov.github.io/murakamis_dream/papers/circle-tree-geometry.pdf.